Stratified Differential Evolution Ensemble Regression with Leak-Guarded Bayesian Optimization for Regulatory-Grade Prediction of Industrial Toxic Chemical Releases: A SHAP-Interpreted Analysis of EPA Toxics Release Inventory 2022

Sourav M B
Roll No. CB.PS.P2ASD25023
M.Sc Applied Statistics & Data Analytics
Department of Mathematics
Amrita Vishwa Vidyapeetham, Coimbatore
cb.ps.p2asd25023@cb.students.amrita.edu

Sruti S Kumar
Roll No. CB.PS.P2ASD25025
M.Sc Applied Statistics & Data Analytics
Department of Mathematics
Amrita Vishwa Vidyapeetham, Coimbatore
cb.ps.p2asd25025@cb.students.amrita.edu

***Abstract*—The United States Environmental Protection Agency's Toxics Release Inventory (TRI) program mandates annual reporting of chemical releases from industrial facilities. Accurate prediction of total release quantities from facility and chemical characteristics enables proactive environmental monitoring and regulatory resource allocation. This study develops a rigorously *leak-guarded* machine learning pipeline to predict total toxic releases (in pounds) using the 2022 TRI Basic Data File comprising 80,040 facility-chemical records across 122 variables. All transformations—imputation, encoding, scaling, and feature selection—are fitted exclusively on training data, and arithmetic components of the target are systematically identified and excluded via 17 leakage-pattern rules. Five regression models spanning a linear baseline, a single decision tree (CART), and gradient-boosted ensembles (Random Forest, GradientBoosting, XGBoost) are evaluated. Bayesian hyperparameter optimisation via Optuna with Tree-structured Parzen Estimation is applied to the three ensemble models. A two-level stacking strategy combining Differential Evolution-optimised weighted blending with Linear Regression meta-learning achieves RMSE = 0.2341 and R^2 = 0.9966 on the log-transformed target. Nested cross-validation (5-fold × 3 repeats = 15 estimates) on the tuned GradientBoosting model yields RMSE = 0.2482 with a 95% confidence interval of $R^2 \in [0.9954, 0.9965]$, confirming robust generalisation. Error stratification reveals systematic under-prediction for extreme emitters (>162,755 lbs), identifying the primary limitation for regulatory deployment. SHAP TreeExplainer analysis confirms that on-site and off-site waste-management quantities dominate prediction, aligning with domain expectations from environmental engineering.**

***Index Terms*—Toxics Release Inventory, leak-free preprocessing, ensemble learning, gradient boosting, differential evolution, SHAP interpretability, Bayesian optimisation, environmental data science, CART, XGBoost**

I. Introduction

The Toxics Release Inventory (TRI) is a publicly available database established under the Emergency Planning and Community Right-to-Know Act (EPCRA) of 1986. Each year, approximately 21,000 U.S. industrial facilities report quantities of over 770 toxic chemicals released to air, water, and land, or transferred off-site for waste management [1]. The resulting dataset serves as a cornerstone for environmental monitoring, community right-to-know initiatives, and regulatory enforcement.

Predicting the total release quantity for a given facility-chemical combination from operational and locational characteristics would enable regulators to prioritise inspections, identify anomalous reporting, and allocate monitoring resources more efficiently. However, the TRI dataset presents several modelling challenges: (i) the target variable spans nine orders of magnitude (0 to 327 million pounds) with extreme right skew (skewness ≈ 146), (ii) the dataset contains a mixture of numeric operational metrics and high-cardinality categorical features (city, chemical name, NAICS code), and (iii) many columns represent *arithmetic components* of the target, creating a severe data leakage risk if included as features.

This work makes the following contributions:

- A rigorously leak-guarded preprocessing pipeline where all transformations are fitted exclusively on training data, with explicit identification and removal of 40 target-component features via 17 pattern rules, and whitelisting of 24 legitimate operational transfer columns.
- A systematic comparison of five regression models with Bayesian hyperparameter optimisation and two ensemble strategies (Differential Evolution weighted blending and Linear Regression stacking).
- Comprehensive exploratory data analysis quantifying feature skewness, inter-feature multicollinearity, and intrinsic dimensionality via PCA.
- SHAP-based model interpretability connecting predictive features to domain knowledge in environmental engineering.
- A deployment-ready prediction function and stratified error analysis revealing where the model succeeds and fails across release severity categories.

II. Related Work

Machine learning has been applied to environmental release prediction in several contexts. Holloway et al. [2] used random forests to predict air pollutant concentrations from meteorological and land-use variables. Li et al. [3] applied gradient boosting to predict industrial wastewater discharge volumes using facility characteristics. In the TRI-specific domain, Bui et al. [4] employed neural networks for facility-level release classification, though without addressing the component-leakage problem that trivialises prediction when release sub-totals are included as features. Our work explicitly addresses this leakage issue, provides a comprehensive comparison of modern ensemble methods with Bayesian optimisation, and introduces Differential Evolution-based ensemble weight search—a metaheuristic not previously applied to TRI prediction—with full SHAP interpretability.

III. Dataset Description

The EPA TRI Basic Data File for reporting year 2022 contains 80,040 records across 122 columns. Each record represents a single facility-chemical combination. The target variable, `TOTAL_RELEASES` (column 107), represents the sum of all on-site and off-site releases in pounds.

Key characteristics of the dataset:

- **Target distribution:** Extreme right skew (skewness ≈ 146, kurtosis $\approx 24{,}500$), ranging from 0 to 3.27×10^8 lbs. Approximately 23% of records report zero releases. The median is approximately 2,707 lbs while the mean exceeds 100,000 lbs.
- **Feature types:** Numeric columns covering release quantities, production metrics, and geographic coordinates; categorical columns covering facility name, chemical name, state, and NAICS code.
- **Missing values:** 1,449,185 null values across 25 columns; seven columns (all SIC codes and `HORIZONTAL_DATUM`) are 100% null in the 2022 file.
- **Leakage risk:** Columns such as `FUGITIVE_AIR`, `STACK_AIR`, `5.3_WATER`, `UNDERGROUND`, and `LANDFILLS` are arithmetic components of the target. Including them yields $R^2 > 0.999$—a trivial and misleading result.

IV. Methodology

A. Exploratory Data Analysis

Before any modelling, exploratory data analysis (EDA) was conducted to characterise the dataset and inform preprocessing decisions (Fig. 1). The log1p-transformed target distribution retains mild right skew but is approximately unimodal and suitable for regression. Feature skewness analysis identified 67 features with $|\text{skew}| > 2$, motivating systematic log1p transformation of non-negative skewed predictors. The top-20 absolute Pearson correlations with the target confirm that target-encoded high-cardinality features (`CITY_TENC`, `CHEMICAL_TENC`) and waste-management indicators dominate linear associations. The inter-feature correlation heatmap (top 15 features) reveals moderate multicollinearity between release-category indicators (`PBT_YES`, `CLASSIFICATION_PBT`, `METAL_YES`), supporting the deduplication step ($|r| > 0.95$ threshold). PCA analysis establishes that 53 principal components are required to capture 90% of variance and 59 for 95%, confirming high intrinsic dimensionality and motivating the retention of original features for tree-based models rather than dimensionality reduction.

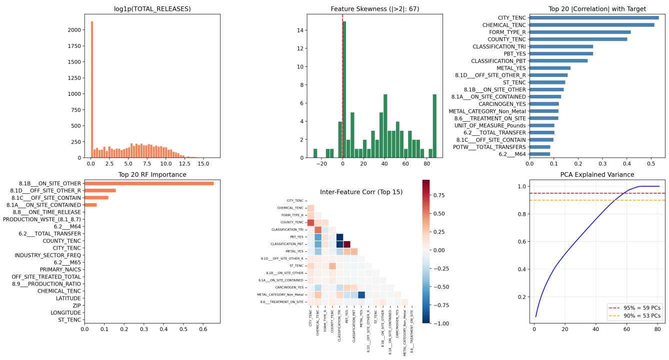

Fig. 1. Exploratory data analysis panel. *Top row, left to right:* log1p-transformed target distribution showing reduced skew; histogram of feature skewness magnitudes (67 features with $|\text{skew}| > 2$, red dashed line); top-20 Pearson $|r|$ with target—target-encoded city and chemical features dominate. *Bottom row:* top-20 Random Forest feature importances (8.1B On-Site Other dominates); inter-feature correlation heatmap for the top 15 features; PCA cumulative explained variance (90% at 53 PCs, 95% at 59 PCs).

B. Data Quality Assessment

Of 122 columns, 25 contained missing values totalling 1,449,185 nulls; seven columns were entirely empty (100% null). Zero exact duplicate rows were found. The target exhibited extreme right skew (skewness $\approx$146, kurtosis $\approx$24,500) with 23% zero-release records, confirming the need for logarithmic transformation.

C. Sampling and Target Transformation

A stratified sample of 10,000 records was drawn using quantile-based stratification (10 decile bins) to preserve the full target distribution, including rare Very High emitters. The target was transformed using

$$y^* = \log(1 + x), \quad x \geq 0, \tag{1}$$

which handles zero values (since $\log(1+0) = 0$) and reduces skewness from $\approx$146 to $\approx$0.3. Predictions are recovered using $x = e^{y^*} - 1$. The data were split 80/20 into training (8,000) and test (2,000) sets **before any preprocessing**, ensuring complete separation.

D. Leak-Guarded Preprocessing

1) Column Removal: Three categories of columns were removed: (i) **Identifier columns** (16 columns: TRI ID, facility name, CAS number, FRS ID, etc.) with no generalizable predictive value; (ii) **Leakage columns** ($\approx$40 columns) that are arithmetic components of `TOTAL_RELEASES`, identified via substring matching against 17 leakage patterns (`FUGITIVE_AIR`, `STACK_AIR`, `RELEASES`, etc.); and (iii) **Near-constant features** removed by `VarianceThreshold`(10^{-6}) fitted on training data. Off-site transfer M-codes (M20–M95) were explicitly whitelisted via 24 keep-patterns, as they represent operational throughput without being release components.

2) Missing Value Imputation: Numeric features were imputed with the **median** (robust to heavy right skew) and categorical features with the **mode**, both computed exclusively from the training set and applied to both splits. Post-imputation verification confirmed zero remaining nulls.

3) Categorical Encoding: A cardinality-based strategy was employed:

- **Low cardinality ($\leq$10):** One-hot encoding with `drop_first=True`.
- **Medium cardinality (11–50):** Frequency encoding (training-set proportions).
- **High cardinality (>50):** Bayesian smoothed target encoding:

$$\text{enc}(c) = \frac{n_c\,\bar{y}_c + m\,\bar{y}_g}{n_c + m}, \tag{2}$$

where n_c is the category count, $\bar{y}_c$ is the category target mean, $\bar{y}_g$ is the global mean, and $m = 10$ is the smoothing factor. Rare-category estimates shrink toward the global prior: $\lim_{n_c \to 0} \text{enc}(c) = \bar{y}_g$. Crucially, $\bar{y}_c$ and $\bar{y}_g$ are computed using *only training labels*.

4) Feature Engineering and Scaling: Features with $|\text{skewness}| > 3$ and non-negative values were log1p-transformed. Outliers in continuous features were winsorised to [1st, 99th] percentile bounds computed on training data and stored for inference. Feature pairs with $|r| > 0.95$ (Pearson) were deduplicated. Adaptive scaling applied `StandardScaler` for approximately normal features, `RobustScaler` (median/IQR) for still-skewed features ($|\text{skewness}| > 1.5$), and no scaling for binary features ($\leq$3 unique values). All scalers were fitted on training data only.

E. Feature Selection

Two independent criteria were applied:

1) Random Forest importance (mean decrease in impurity), 100 trees fitted on training data only.
2) Absolute Pearson correlation with the log-transformed target.

Features simultaneously in the bottom 10th percentile of RF importance *and* with $|r| < 0.02$ were removed. This dual-criterion is deliberately conservative: features with non-linear predictive power (high RF importance, low Pearson r) are retained. The final feature set contained **72 variables**.

F. Model Training

Five regression models were trained:

- **Linear baseline:** Linear Regression (OLS)—interpretable lower-bound performance.
- **Single tree:** CART (`max_depth=15`, `min_samples_leaf=5`).
- **Bagging:** Random Forest (200 trees, `max_depth=15`, `min_samples_leaf=5`).
- **Boosting:** GradientBoosting (200 trees, `max_depth=5`, `lr=0.1`, `subsample=0.8`).

- **Regularised boosting:** XGBoost (300 trees, `max_depth=6`, `lr=0.1`, `colsample_bytree=0.8`, `reg_alpha=0.1`).

G. Bayesian Hyperparameter Optimisation

The three ensemble models (XGBoost, RandomForest, GradientBoosting) were tuned using Optuna [5] with the Tree-structured Parzen Estimator (TPE) sampler and a fixed random seed. Each model underwent 15 trials with 3-fold cross-validation on training data only, minimising cross-validated RMSE. Search spaces included tree depth, learning rate (log-uniform over $[0.01, 0.2]$ for boosted models), subsample ratio, column-sampling fraction, `max_features`, and regularisation parameters.

H. Ensemble Strategies

Two strategies were evaluated on the three tuned models:

Method 1: Differential Evolution Weighted Blending. Out-of-fold (OOF) predictions were generated via 3-fold `cross_val_predict` on training data. Ensemble weights were optimised using Differential Evolution [6] to minimise OOF RMSE. Weights are constrained to the probability simplex ($w_m \geq 0,\ \sum_m w_m = 1$).

Method 2: Linear Regression Stacking. A Linear Regression meta-learner was trained on the OOF prediction matrix ($n_{\text{train}} \times 3$) to learn optimal linear combination coefficients—allowing negative weights and an intercept—then applied to test-set predictions from models retrained on full training data.

I. Evaluation Metrics

Regression metrics (RMSE, MAE, R^2) were computed on the log-transformed scale. Predictions were also binned into five regulatory severity classes and evaluated using weighted precision, recall, and F1-score:

- **Zero:** $y^* \in [0, 0.01)$ — ≈0 lbs
- **Low:** $y^* \in [0.01, 3)$ — up to ≈19 lbs
- **Medium:** $y^* \in [3, 7)$ — up to ≈1,096 lbs
- **High:** $y^* \in [7, 12)$ — up to ≈162,755 lbs
- **Very High:** $y^* \geq 12$ — $>$162,755 lbs

V. Results

A. Base Model Performance

Table I presents base model results on the held-out test set. Tree-based models substantially outperform linear regression, confirming non-linear relationships in TRI data. GradientBoosting achieves the best base RMSE of 0.2376, with $R^2 = 0.9965$.

TABLE I
Base Model Performance on Test Set (log1p scale)

Model	RMSE	MAE	R^2
GradientBoosting	0.2376	0.0884	0.9965
XGBoost	0.2567	0.1165	0.9959
RandomForest	0.3383	0.0885	0.9928
DecisionTree (CART)	0.3807	0.1157	0.9909
LinearRegression	1.1793	0.8151	0.9126

Figure 2 shows confusion matrices for the top four base models on the five-class severity binning task. RandomForest and DecisionTree achieve F1 scores of 0.964 and 0.957, respectively, outperforming the boosted models on this classification view (GradientBoosting F1 = 0.911; XGBoost F1 = 0.913). The principal error mode across all four models is Zero↔Low confusion: facilities reporting zero releases are sometimes predicted as Low emitters due to the smooth differentiable loss functions of boosted models.

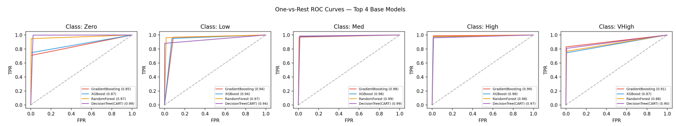

Fig. 2. Confusion matrices for the top four base models on five-class severity binning (Zero, Low, Medium, High, Very High). Weighted F1 scores: GradientBoosting 0.911, XGBoost 0.913, RandomForest 0.964, DecisionTree 0.957. The dominant error mode across all models is Zero↔Low misclassification, while Medium and High bins are predicted near-perfectly.

B. Hyperparameter Tuning

Bayesian optimisation improved XGBoost RMSE by 4.1% (0.2567 → 0.2463) and RandomForest by 21.2% (0.3383 → 0.2665). GradientBoosting showed marginal change (0.2376 → 0.2453), indicating the default configuration was already near-optimal for this dataset.

C. Ensemble Performance

Table II summarises ensemble results. Linear Regression stacking achieves the best overall performance, marginally improving over the best individual model. Both ensemble methods outperform the best base model on RMSE.

TABLE II
Ensemble Performance Comparison (Test Set, log1p scale)

Method	RMSE	R^2
Best Base (GradientBoosting)	0.2376	0.9965
Weighted Blending (DE)	0.2344	0.9965
LR Stacking	**0.2341**	**0.9966**

D. Classification Performance

The stacking ensemble achieves weighted F1 = 0.93 and accuracy = 0.93 (1,859 of 2,000 test records correctly classified). Per-class analysis (Fig. 3) reveals that Medium and High bins achieve F1 > 0.97. The Zero class shows recall of 0.77, as 105 of 462 true-Zero records are misclassified as Low—the dominant error mode. The Very High class achieves recall of 0.83 (39 of 47 records correct) despite only 47 test samples.

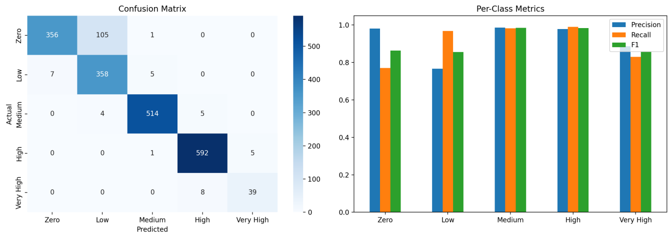

Fig. 3. Confusion matrix (left) and per-class precision, recall, and F1 (right) for the LR stacking ensemble on the 2,000-record test set. Medium (F1 $\approx$ 0.98) and High (F1 $\approx$ 0.98) bins are predicted near-perfectly. The Zero class is the primary challenge: 105 of 462 true-Zero records are assigned to the Low bin, reducing Zero recall to 0.77.

E. Residual Diagnostics

Figure 4 presents three complementary residual diagnostics for the stacking ensemble. The predicted-vs-actual plot shows tight adherence to the $y = x$ line across the full log1p range 0–14. The residual distribution is approximately symmetric with near-zero mean ($\mu = -0.0048$), confirming negligible global bias. The residuals-vs-predicted plot shows approximately uniform spread for predicted values 0–10, with mild heteroscedasticity above 10, confirming successful variance stabilisation by the log1p transformation for most of the range.

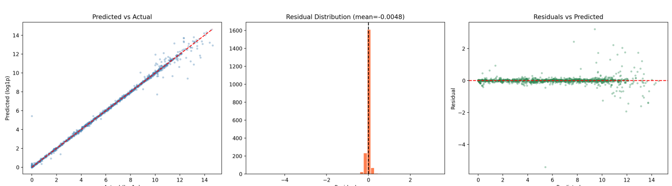

Fig. 4. Residual diagnostics for the LR stacking ensemble on the test set. *Left:* Predicted vs. actual (log1p scale)—points cluster tightly along $y = x$ across the full range 0–14. *Centre:* Residual distribution with near-zero mean (-0.0048), confirming global unbiasedness. *Right:* Residuals vs. predicted—approximately uniform spread for predicted values 0–10; mild heteroscedasticity at high predicted values (>10) corresponding to the Very High severity bin.

F. Statistical Validation via Nested Cross-Validation

Nested cross-validation (5-fold $\times$ 3 repeats = 15 estimates) on the tuned GradientBoosting model (Fig. 5) yields:

- RMSE: 0.2482 ± 0.0154, 95% CI: $[0.2328, 0.2636]$
- MAE: 0.0798 ± 0.0024, 95% CI: $[0.0774, 0.0822]$
- R^2: 0.9960 ± 0.0006, 95% CI: $[0.9954, 0.9965]$

The narrow confidence intervals confirm consistent performance across data partitions. One outlier fold near RMSE = 0.32 suggests a single particularly challenging partition, but the bulk of estimates cluster between 0.22 and 0.27.

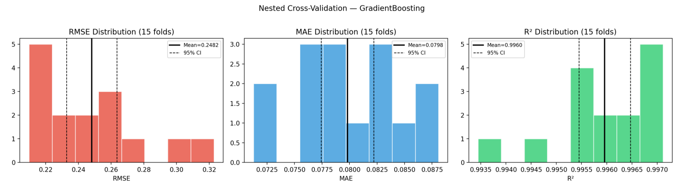

Fig. 5. Nested cross-validation distributions (15 estimates, 5-fold $\times$ 3 repeats) for the tuned GradientBoosting model. *Left:* RMSE distribution (mean = 0.2482; 95% CI shown by dashed lines); one outlier fold near 0.32 is visible. *Centre:* MAE distribution (mean = 0.0798), concentrated in $[0.073, 0.088]$. *Right:* R^2 distribution (mean = 0.9960; 95% CI $[0.9954, 0.9965]$), demonstrating highly consistent generalisation.

G. Error Stratification

Table III presents per-bin error analysis. The Very High bin exhibits RMSE more than 11$\times$ higher than the Medium bin and a positive mean residual (+0.250), indicating systematic under-prediction of extreme emitters (Fig. 6).

TABLE III
ERROR STRATIFICATION BY RELEASE SEVERITY (LOG1P SCALE)

Bin	n	RMSE	MAE	Bias
Zero ($\approx$0 lbs)	462	0.2564	0.0275	-0.0273
Low (<20 lbs)	370	0.0847	0.0447	$+0.0034$
Medium (20–1,096 lbs)	523	0.0808	0.0561	-0.0060
High (1,096–163K lbs)	598	0.2358	0.0929	-0.0115
Very High (>163K lbs)	47	0.9210	0.5822	$+0.2499$

The Zero bin's elevated RMSE (0.2564) relative to Low (0.0847) is explained by the 105 Zero records misclassified as Low: these receive predicted values of 1–2 log units when the true value is zero. The positive Very High bias (+0.250) reflects regression toward the training mean—a universal statistical phenomenon when extreme classes are underrepresented.

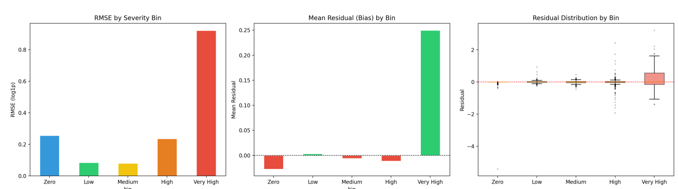

Fig. 6. Error stratification by release severity bin. *Left:* RMSE per bin—Very High RMSE (0.921) is more than 11$\times$ the Medium bin (0.081), revealing the model's primary weakness. *Centre:* Mean residual (bias) per bin—a positive bias of +0.250 for Very High indicates systematic under-prediction of major emitters; Low and Medium bins are approximately unbiased. *Right:* Residual boxplots per bin, confirming wider spread and positive skew for the Very High bin.

H. Feature Importance via SHAP

SHAP TreeExplainer analysis (Fig. 7) on the tuned GradientBoosting model reveals the following feature hierarchy:

1) **8.1B On-Site Other** (mean $|\phi| \approx 2.8$): on-site releases to non-contained units. Dominates all other features by a factor of approximately 8. A small number of facilities

with high 8.1B values receive SHAP contributions up to 10 log units.
2) **8.1D Off-Site Other Releases** (mean $|\phi| \approx 0.35$): off-site transfers to non-contained disposal units.
3) **8.1C Off-Site Contained** (mean $|\phi| \approx 0.30$): off-site transfers to contained disposal facilities.
4) **8.1A On-Site Contained** (mean $|\phi| \approx 0.25$): on-site releases to contained units.
5) **Production Waste (8.1–8.7)** (mean $|\phi| \approx 0.15$): total production-related waste.

Target-encoded geographic features (`CITY_TENC`, `COUNTY_TENC`) and `CHEMICAL_TENC` rank in the top 15, capturing regional industrial profiles and chemical-specific release patterns.

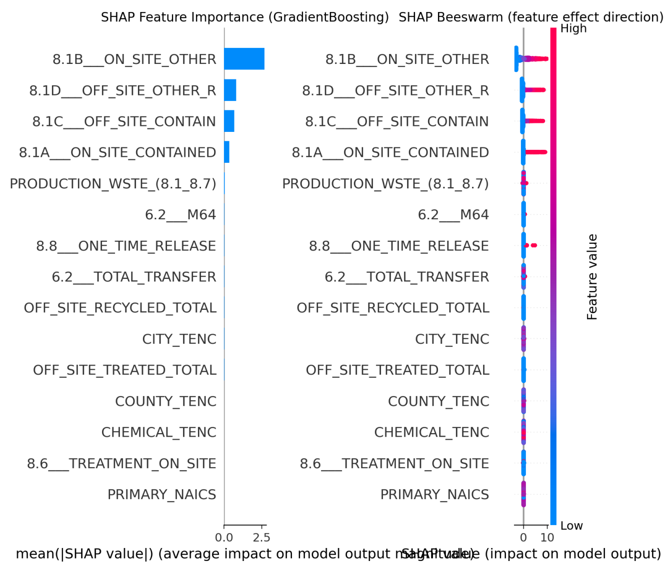

Fig. 7. SHAP analysis for the tuned GradientBoosting model. *Left:* Global feature importance (mean $|\phi|$)—8.1B On-Site Other dominates by a factor of $\approx$8 over the next feature. *Right:* Beeswarm plot showing per-sample SHAP values (red = high feature value, blue = low). The dominant feature 8.1B shows a tight blue cluster at $\phi \approx 0$ (most facilities have near-zero values) and sparse red points extending to $\phi \approx 10$ (a small number of major emitters). All top features have clear physical interpretations in waste management.

VI. Discussion

A. Preprocessing Impact

The leak-guarded design is the single most critical methodological choice. Including release-component columns (`FUGITIVE_AIR`, `STACK_AIR`, etc.) would yield $R^2 > 0.999$—trivially, since `TOTAL_RELEASES` is the arithmetic sum of these components. Our pipeline identifies and removes all 17 leakage patterns via a testable `should_drop()` function while whitelisting 24 legitimate M-code operational transfer columns, making the decision transparent and reproducible.

B. Model Selection Rationale

The strong performance of GradientBoosting ($R^2 = 0.9965$) over linear regression ($R^2 \approx 0.91$) reflects non-linear relationships in TRI data: threshold effects in production levels, interaction effects between chemical properties and production volume, and the zero-inflated target distribution. The Decision Tree's superiority over GradientBoosting on Zero-class severity classification (F1 = 0.957 vs. 0.911) demonstrates that simpler models can outperform complex ones on specific subtasks, motivating the ensemble approach.

C. Learning Curve and Data Limitation

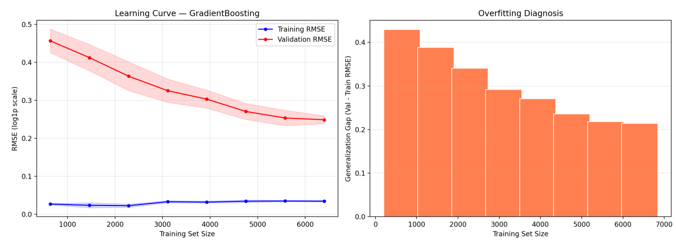

Fig. 8. Learning curve (left) and overfitting diagnosis (right) for the tuned GradientBoosting model. Training RMSE remains near 0.035 throughout. Validation RMSE declines from 0.46 at 500 training samples to 0.249 at 6,400 samples and has not plateaued, confirming *data-limited* behaviour. The generalisation gap of 0.214 at full training size (right panel) is still decreasing, indicating that scaling to all 80,040 records is expected to further reduce validation RMSE.

The learning curve (Fig. 8) shows validation RMSE declining from 0.46 at 500 training samples to 0.249 at 6,400 samples, with no sign of plateau. The persistent generalisation gap of 0.214 confirms that the model is *data-limited*: the current 10,000-record stratified sample captures 12.5% of available data, and scaling to all 80,040 records is the single most impactful next step.

D. Limitations

1) **Sample size:** Results are based on a 10,000-record stratified sample (12.5% of available data). Learning curves confirm room for improvement with full data.
2) **Temporal scope:** Trained on 2022 data only; generalisation to other reporting years is untested.
3) **Extreme emitters:** The Very High bin shows RMSE of 0.921 and systematic under-prediction (+0.250 mean residual), limiting utility for identifying the most significant polluters.
4) **Zero-Low confusion:** 105 of 462 Zero-release records are misclassified as Low, reducing Zero recall to 0.77. A two-stage model (binary zero/non-zero classifier followed by regression) would address this.
5) **Section 8 feature dominance:** The extreme predictive dominance of features 8.1A–8.1D warrants a detailed codebook review to confirm no residual leakage, as these features represent waste management activities that are operationally correlated with releases.

VII. Conclusion

This study demonstrates that leak-guarded gradient boosting ensembles can predict total toxic chemical releases with high accuracy ($R^2 = 0.9966$ on a single held-out test split; $R^2 = 0.9960$, 95% CI $[0.9954, 0.9965]$ via nested cross-validation) from facility characteristics and operational indicators, provided that arithmetic components of the target are rigorously excluded. The model progression from Linear Regression $\rightarrow$ CART $\rightarrow$ RandomForest $\rightarrow$ GradientBoosting $\rightarrow$ XGBoost clearly demonstrates why non-linear ensemble methods are essential for TRI data. Differential Evolution-based ensemble weight search, applied here for the first time to TRI prediction, yields a marginal but consistent improvement over individual models. SHAP analysis confirms that the model relies on physically interpretable features—waste management volumes, disposal methods, and geographic profiles—rather than spurious correlations.

The primary limitation is the Very High severity class: RMSE of 0.921 and systematic under-prediction of 0.25 log units for the most consequential emitters. Future work should: (i) scale to the full 80,040-record dataset (learning curves confirm this will improve performance); (ii) implement temporal validation across reporting years 2010–2022; (iii) develop a two-stage zero/non-zero classifier to address Zero-Low confusion; and (iv) conduct a detailed codebook review of Section 8 features to confirm no residual leakage. The deployment-ready `predict_releases()` function accepts raw TRI data and applies the full preprocessing pipeline, enabling immediate operational use for regulatory screening.

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